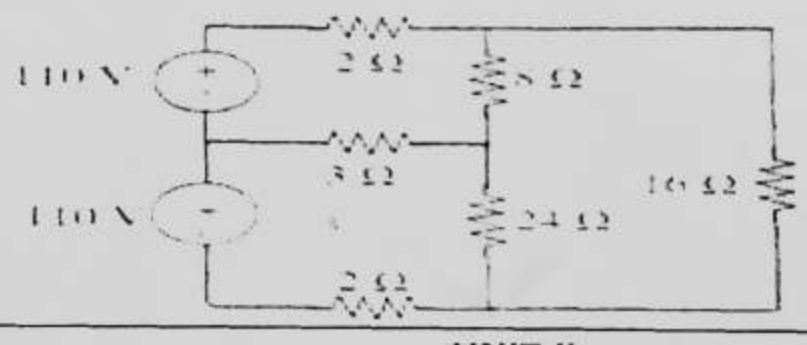
B.Tech ECE/ECE-AI IIIrd SEM
(November-December, 2023)

Subject Code: BEC 205 | **Subject: Network analysis and synthesis**

Time :3 Hours | **Maximum Marks :60**

Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		**(2.5*8=20)**	
	(a) Derive hybrid parameters in terms of Z parameter.		
	(b) How do you find Thevinin's resistance?		
	(c) A heater elements draws 20 A when connected to a 230-V line. How long does it take to consume 60 kJ?		
	(d) Differentiate analysis and synthesis and specify their importance with relevant expression.		
	(e) A series RL circuit has R = 10 KΩ, L = 10 Mh and C = 1 μF. Find the transfer function of the circuit.		
	(f) Derive the expression of Maximum power for the network under maximum power transfer condition.		
	(g) Derive an expression for the decay current in an RC circuit excited by a unit step voltage. What is the time constant of the circuit?		
	(h) Discuss Telegan's theorem.		
	UNIT-I		
Q2	If two circuits X and Y are to be connected in cascade. Give the two port parameters of the combination with diagram of cascade combination	**(10)**	
Q3	Using nodal analysis find all branch currents for the following circuit 	**(10)**	
	UNIT-II		
Q4	A series RLC circuit with R = 2 ohm, L = 1 H and C = 0.5 Farad with the applied voltage V (t) = sin t. Find i (t) if the switch is closed at t= 0. Use Laplace transform method.	**(10)**	
Q5	A voltage pulse of magnitude 8 Volts and duration 2 seconds extending from t=2 seconds to t=4 seconds is applied to a series RL circuit. Obtain the expression for the current i(t).	**(10)**	
	UNIT-III		
Q6	Explain Hybrid and Transmission parameters with examples.	**(10)**	
Q7	Explain the concept of interconnection of Two-Port networks. Show that $H=H_a + H_b$ for two Two-Port networks, when these are connected in series-parallel connection.	**(10)**	
	UNIT-IV		
Q8	How is a physical network realized by considering the driving point admittance function. Discuss.	**(10)**	
Q9	Find the first and second Cauer forms of LC networks for the given driving point impedance function $Z(s) = (s^4 + 10s^2 + 9) / (s^3 + 4s)$	**(10)**	